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Kyle Tran

MECHANICAL ENGINEERING · ROBOTICS MINOR · NORTHEASTERN '29

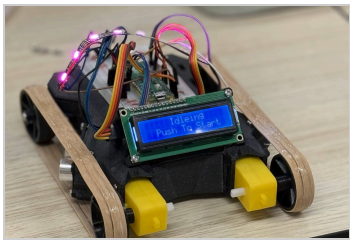
Seeking a Summer 2027 co-op in hardware and robotics. I build robots, autonomous vehicles, and embedded systems — drawn in CAD, machined and printed, wired and coded, then tested until they hold up: two years of competitive combat robotics, and an autonomous survey boat built from scratch.

4th NATIONALLY, 2024	**3.87** GPA	**10** ROBOTS BUILT	**16** TEAM PODIUMS


D.R.E.A.M.S v1 on the water


Creators' Showcase, NU London


Combat robots — Team DHM


Maze-navigating robot

PROJECTS

Autonomous Environmental Survey Boat — D.R.E.A.M.S

Jan 2026 – Present

A twin-hull autonomous survey boat, co-founded with Matthew Hong. V1, built in one semester in London, navigated to GPS waypoints on its own at St Katharine Docks and winched a sensor pod to the bottom to sample. V2 added a Raspberry Pi 5, stereo vision and our first custom PCB, but bench tests exposed flaws in its motor control, power and safety. V3, in build, pairs a Pi 5 with a Teensy 4.1 on a custom 4-layer carrier board. I lead the hardware; the software was built with AI coding agents.

SolidWorks · ASA + fiberglass hull · KiCad 4-layer PCB · Pi 5 + Teensy 4.1 · CAN · VESC · RS-485 · YOLOv8 · GPS/IMU · 4G

Combat Robotics — Team DHM

2023 – 2025

Designed and built around ten robots that competed across five weight classes — weapon and drivetrain design, titanium and steel armor, composite layup, and iteration driven by what broke in the arena. 4th nationally in Plastic Antweight (2024), U.S. rankings of top-15, top-30, and top-35, and 4 firsts and 8 seconds at regional events; the club took 16 podiums over 5 events. The newest Achilles walks on a single-cam shuffler drive I designed.

Onshape · CNC-machined aluminum (outsourced) · laser-cut Ti / steel · brushless drive · FDM · DFM to a weight limit

Adaptive Compliant Robotic Arm

2026 – in design

A five-axis arm with a ±0.25° static target and a compliant TPU gripper that reads its own grip force. The base, shoulder and elbow run cycloidal gearboxes (14:1 to 22:1) with output-side magnetic encoders, and all three pass every check in a gearbox simulator I specified and built with AI coding agents. Nothing is built yet: a printed prototype comes first, then 7075 discs laser-cut by SendCutSend and finished on a Bantam PCB mill that I run.

Cycloidal drives · gearbox simulator · closed-loop steppers · output-side encoders · TPU gripper · tolerance budgeting

Autonomous Maze-Navigating Robot

Oct – Dec 2025

A 3D-printed tank-drive robot, built with Matthew Hong for the Cornerstone of Engineering course at Northeastern, that completed all three challenge courses. I did the CAD, the printing and the MicroPython firmware: proportional steering off two angled ultrasonic sensors, with asyncio so the LCD and LEDs never stall the motors.

Onshape · 3D printing · MicroPython · asyncio · RP2040 · ultrasonic + IR · I2C LCD

Miscellaneous Builds

2023 – Present

10+ custom ceramics tools designed in CAD, printed, and open-sourced for anyone to print; 3D-printed RC aircraft with carbon-fiber spars; a full sewing machine reverse-engineered in CAD; and printed climbing hardware.

CAD to FDM · published on MakerWorld · github.com/ka1e27

Experience & Skills

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EDUCATION

B.S. Mechanical Engineering · Minor in Robotics

Sep 2025 – May 2029 (exp.) · GPA 3.87

Northeastern University · Boston, MA

Cornerstone of Engineering I · Statics · Thermodynamics · Materials Science · Probability & Statistics

LEADERSHIP EXPERIENCE

President

Oct 2023 – Jun 2025

Combat Robots Club, Burlingame

Ran workshops on CAD, soldering, physics, materials, and 3D printing; handled sponsorships and club finances. The team placed on the podium 16 times — 4 firsts, 7 seconds, 5 thirds — across 5 events.

Ceramic Tools R&D

Oct 2023 – May 2025

Burlingame High School · Ceramics Program

Modeled, printed, and published 10+ ceramics tools — trimming tools, ribs, wire guides, stamps — free on MakerWorld.

Secretary · Co-founder

Jan 2024 – May 2025

Burlingame Aeronautics Club

Helped found the club and taught the physics of flight — aerodynamics, lift, drag — then put it into practice designing 3D-printed RC aircraft.

Subsystem Lead

Sep 2021 – May 2024

Iron Panthers FRC #5026

Designed the spindexer mechanism for the competition robot, working with 10+ team members and 5 professional mentors through design reviews and manufacturing.

SKILLS

DESIGN & CAD

SolidWorks · Onshape · Fusion 360 · mechanism design · DFM

ELECTRONICS & FIRMWARE

KiCad · custom PCB · Raspberry Pi · RP2040 · MicroPython · soldering

SOFTWARE & AI

Python/asyncio · MATLAB · Java · AI coding agents · simulation-first development

FABRICATION

FDM printing · composite layup (FG/CF) · laser cutter · PCB mill · beginner: 2.5D CNC, CNC router, waterjet

ROBOTICS & AUTONOMY

Stereo camera rigs · GPS + IMU fusion · VESC drive · 4G telemetry · state machines

LEADERSHIP

Club president · workshops · sponsorships · mentoring

AWARDS

4th	Robot Combat League National Championship	1 lb plastic antweight · 2024
1st	CROS — March Madness	1 lb antweight · 2024
1st	BCRC Bell Brawls IV	3 lb beetleweight · 2024
1st	Spartan Bell Slugfest	1 lb plastic antweight · 2024
1st	NorCal Robotics Expo	1 lb plastic antweight · 2023
2nd	Eight further second-place finishes	ant / beetleweight · 2024